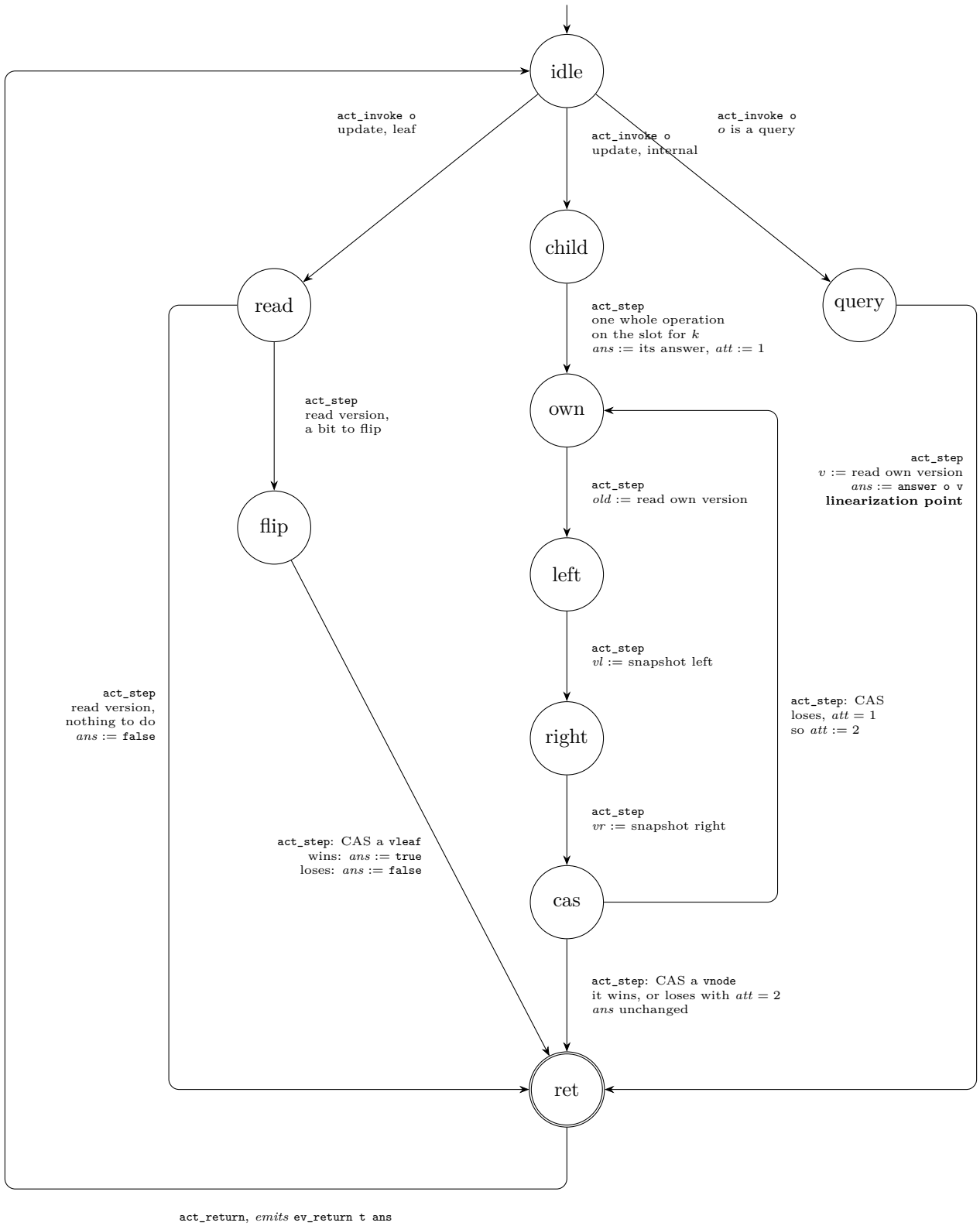




Key. Each circle is a value of CPC, and the italic names are the local variables it carries: *o* the operation, *ans* the answer owed, *att* which of the two refresh attempts this is, and *old*, *vl*, *vr* the three versions a refresh reads. Every arrow is one atomic step, one read or one CAS; a local test costs no memory access and is decided inside the step that reaches it. All three arrows out of **idle** emit **ev_invoke t o** and are taken only when **op_ok i j o** holds. Every CAS compares tags rather than shape, and the two that win consume a tag from **cfresh**. The four arrows **own**, **left**, **right**, **cas** are one **refresh**, and going round them twice is the double refresh. Any action not drawn is a stutter, leaving the configuration alone and emitting nothing: a disallowed **act_invoke**, an **act_return** anywhere but **ret**, and **act_step** at **idle** or at **ret**. Which of the three paths a thread takes is settled by the arrow it leaves **idle** on and cannot change afterwards, so no later state retests whether the component is a leaf; that is the lemma **pc_shape_run**.